

Effects of Metal Binding on Protein Structure

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I. INTRODUCTION	I
II. ANION BINDING PROTEINS	5
III. UNIVALENT CATIONS BINDING PROTEINS	6
IV. CALCIUM BINDING PROTEINS	9
V. MANGANESE AND MAGNESIUM BINDING PROTEINS	16
VI. ZINC BINDING PROTEINS	20
VII. CONCLUSION	24
VIII. REFERENCES	26

I. INTRODUCTION

A distinction has been made between 'metalloproteins' and 'metal-protein complexes'. The former exhibit high metal-ligand stability constants (the metal is not removed during the isolation of the protein), while the latter bind metal ions only loosely (Vallee, 1955). Actually, all proteins can be considered as existing as metal-protein complexes. Selectivity, however, among proteins in their tendency to combine with inorganic ions (both with regard to the type of ion as well as number of ions) suggests that particular configurations produce specific reactive sites. In turn, binding of ions to these sites alters the electronic and consequently the steric conformation of the protein. Thus, effects on macromolecular conformation may not only be due to structural changes of the solvent induced by ions, which modifies solvent-macromole

interaction, but also to specific ligation of the ions to the protein which alters local molecular arrangement.

Data obtained for some proteins by measurements with the perm-selective membrane electrode are listed in Table 1. Intrinsic association constants of 10 to 20 would indicate that the free energies for such binding are only two or three times the thermal energy. The $-\Delta G$ for inorganic binding is small (-3000 cal/mole) since only electrostatic interactions are involved. This energy represents not merely the interaction of the protein with the ion but also the interactions of the protein with solvent and of the ion with solvent which have to be readjusted whenever ligation of the ion occurs. (Metal ions are always hydrated to some extent.) A decrease of a few hundred calories per mole in the forces involved in maintaining a certain conformation of the protein molecule should be enough to induce a local structural arrangement or even a cooperative transition to a different structure.

Table 1 also shows that myosin A binds both Na^+ and K^+ ions. The association constants for the ligation of these ions to this non-globular protein are considerably higher than those for the globular proteins listed. (The number of ions, however, bound by the myosin is not exceptionally large if expressed on a weight rather than molecular basis.)

In most instances, a variation of n and K with pH (not shown in the table but documented in the original papers) suggests the possibility of competition of the metal ions with hydrogen ions for the binding site. At a physiological pH, carboxyl groups, however, would be completely deprotonated and hence no competition should occur between protons and metal ions for this group. Similarly, the stability of imidazole complexes is only slightly affected by pH changes in that range. On the other hand, metals ligated to amino groups could be strongly influenced by change in pH. The $-\text{SH}$ sidechain and the phenolic sidechain also exhibit strong competition by protons and would bind Cu^{2+} and Fe^{3+} , respectively, i.e. metals which coordinate forcefully.†

The ions of transition metals are dense and exhibit relatively small atomic radii. Besides electrostatic attractions they display electromagnetic ones and hence these elements possess a greater affinity for proteins than do Na^+ or K^+ . Thus Co^{2+} ions are bound so tightly to serum albumin that the metal is not released upon dialysis or ammonium sulphate fractionation, i.e. by definition this protein can be considered

† Because of the uniquely high energy binding of iron and copper ions the discussion of the interaction of these two ions with proteins is reserved for a separate review.

TABLE 1. Overall association constants and affinities of the first ion to various proteins

Protein	pH	Ion	$K_1 (= nK)^\dagger$	n	$-\Delta G_1 = RT \ln K_1$ (cal)
Insulin [†]	3.4	Cl ⁻	150	6	3000
ATP creatine phosphotransferase*	5.7	Cl ⁻	210	9	3150
	6.7	K ⁺	180	12	3070
Myoglobin§	7.1	SO ₄ ²⁻	45	—	2250
	9.4	K ⁺	60	3	2400
α-Chymotrypsin¶	6.7	Cl ⁻	50	3	2300
	6.7	SO ₄ ²⁻	90	3	2650
	6.8	Na ⁺	190	11	3100
	6.8	Ca ²⁺	290	17	3950
Ribonuclease††	4.5	Cl ⁻	1500	2	4400
	4.5	SO ₄ ²⁻	1.2×10^4	2	5550
Myosin A‡‡	7.7	K ⁺	4210	43	5000
	7.7	Na ⁺	9450	42	5500
Serum albumin**	7.1	Co ²⁺	1.7×10^4	25	5800
Serum albumin§§	7.0	Mn ²⁺	2.7×10^4	6	6100

† If protein (P) contains maximum number (n) of binding sites per mole capable of reacting with ion (I), then the equilibrium constant (overall association constant) (K_1) for the total reaction of P with the first ion can be expressed as:

$$K_1 = \frac{PI}{(P)(I)} = nK,$$

where K is the intrinsic (or apparent molar) association constant, that is, the equilibrium constant for one site with the first ion, and

$$\bar{V}/c = Kn - K\bar{V}$$

$\bar{V}$ represents the moles of ion bound per mole of protein and c the concentration of free ion. A plot of $\bar{V}/c$ versus $\bar{V}$ yields Kn as intercept on the $\bar{V}/c$ axis and the intercept on the $\bar{V}$ axis is equal to n .

† Fredericq (1956).

* Floyd & Friedberg (1966).

§ Friedberg & Emiola (1968).

¶ Friedberg & Bose (1969).

†† Saroff & Carroll (1962).

‡‡ Lewis & Saroff (1957).

** Nandedkar, Basu & Friedberg (1973).

§§ Nandedkar, Nurse & Friedberg (1973).

a metalloprotein. Albumin, however, unique in its capacity to bind organic molecules such as dyes and L-tryptophan also displays unusual attractive forces towards Mn²⁺, etc., besides towards Co²⁺ (see Table 1).

Of course, metalloproteins such as transferrin may hold ions with such great strength that binding constants may not be measurable and the complexing may be declared as irreversible thermodynamically. In

such instances, one must ask how the biological transfer of the metal from the protein carrier to a suitable acceptor can be achieved.

The structural rearrangement induced by ion binding in a protein should confer stability. The introduction of such stability is a complex phenomenon since ions interact with the protein at various sites, probably restricting mobility of portions of the protein molecule by cross-linking charged groups. Lovrien & Sturtevant (1971) measured the enthalpy of ligating iodide to deionized serum albumin by calorimetry. When combined with free energy data, the enthalpy ($-17.4 \text{ kcal mol}^{-1}$) for the strong binding process (there is one strong binding site ($n = 1$) and several weaker ones ($n = 8$ and $n = 18$) yields $-42 \text{ cal deg}^{-1} \text{ mol}^{-1}$ for the standard entropy of binding, a rather large negative value. This value cannot be explained merely in terms of changes in ionic solvation and thus suggests that the attachment of the ion curtails conformational freedom.

For fairness, it should also be mentioned that in the case of the attachment of cupric ions to serum albumin, the enthalpy changes range from $+4$ to $+2.2 \text{ kcal mol}^{-1}$ and the entropy changes from $+31$ to $+17 \text{ cal deg}^{-1} \text{ mol}^{-1}$ for the first to the tenth bound ion, respectively (there are four stronger ($K = 1370$) and twenty weaker ($K = 44$) binding sites for this metal). Utilizing hydrogen-deuterium exchange studies, conformational changes cannot be observed until at least 10 moles of copper are ligated. Thereafter, a substantial alteration is registered (Reynolds, Burkhead & Mueller, 1973). Hydrogen-deuterium exchange, however, does not provide an index sensitive enough to monitor local structural rearrangements. And Klotz (1967) explains the positive change in entropy as due to the simultaneous release of frozen water from the protein and from the copper accompanying the formation of the protein- Cu^{2+} complex.

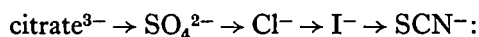
The paramount questions – what conformational changes are induced by the binding process and where precisely in the protein molecule are the ligated ions located – are in focus now, and some of the answers so far given in the literature are summarized below.

While emphasis in this review is placed primarily on data for binding cations, ligation of anions should also be effective in triggering conformational changes.

II. ANION BINDING PROTEINS

Phosphoglucomutase shows a difference-spectrum in the 260–300 nm region when a solution of this enzyme containing 0.01 M divalent anions (SO_4^{2-} or HPO_4^{2-}) is compared to the one containing monovalent anions (Cl^-), indicating that the environment of tryptophan residues is involved (Yankeelov & Koshland, 1965).

X-ray diffraction data for apoenzyme dogfish M_4 lactate dehydrogenase suggest existence of two exchangeable anion binding sites. One sulphate ion is held at the active centre of the apoenzyme (where pyruvate sits in the LDH-NAD-pyruvate complex) by histidine 195 and arginine 171. In the ternary complex, due to a large change in conformation, arginine 109 and the nicotinamide become also available as ligands. The other sulphate, held by arginine 173, sits between two subunits (the second negative charge of this anion might be neutralized by a cation from the solvent) and it is possible that the intersubunit binding site is responsible for the stabilization of the tetramer which takes place in the presence of sulphate ions. (It is suggested, however, that pyruvate is bound in the ternary complex at this inter-subunit binding site rather than a sulphate ion Adams, Liljas & Rossman, 1973). Sulphate ions inhibit hybridization of the enzyme while halides and thiocyanate promote it. In proceeding from 'structure makers' to 'structure breakers':



i.e. dissociating power increases with entropy of hydration (Jaenicke, Koberstein & Tenscher, 1971).

The transferrin molecule contains two specific binding sites for Fe^{3+} . Usually, for each metal ion ligated, a bicarbonate ion attaches concomitantly. Nitrate cannot, but certain compounds having two or more carboxyl groups (e.g. oxalate) can substitute for the bicarbonate ion. Neither metal ion nor anion is tightly bound without the presence of the other. (Since in the absence of anion, metal binding is very weak, it is possible that release of anion proceeds biological transfer of Fe^{3+} from transferrin.) In all likelihood, the bicarbonate is not held directly by the metal, since both the copper and the iron protein complexes ligate the anion and the two metals have different co-ordination properties. (Also, there is no evidence that an extra water molecule is attached to the metal when no anion is bound.) Thus, one may postulate that bicarbonate induces a conformational change which strengthens iron

binding capacity and similarly, the iron enhances bicarbonate binding capacity (Aisen *et al.* 1967). Exchangeability studies of bicarbonate by monoferric and diferric transferrin suggest that interaction between two specific anion-binding sites of the proteins occurs when each of these two sites is occupied (Aisen *et al.* 1973).

2,3-Diphosphoglycerate, which lowers the oxygen affinity of haemoglobin is bound to the deoxyform (1 mole per mole of protein) with a K of 1.3×10^5 (at neutral pH) in such a way that its phosphates are close to the *N*-terminal of the β -chains. Upon ligation, the organic phosphate might facilitate stabilization of the quaternary structure of the protein (Perutz, 1970). The affinity of haemoglobin for oxygen decreases with increasing NaCl concentration reflecting preferential binding of the salt by the deoxy form. (2,3-diphosphoglycerate, however, displays an efficacy about three orders of magnitude greater than that of the NaCl.) It is quite possible that Cl^- ions bind at the same point where the 2,3-diphosphoglycerate attaches (Benesch, Benesch & Yu, 1969), i.e. they would compete with the organic ions for the same binding sites even though their association constants are probably of the order of 10^2 or $10^3 (\text{M}^{-1})$.

Horse liver alcohol dehydrogenase, a dimeric molecule, binds 2 moles of NADH to its two active sites and 3.1–4.3 g-atoms Zn^{2+} per mole of enzyme. V_{max} of the dehydrogenase, with ethanol and NAD^+ as substrates, is increased by one third if 0.05 M- Cl^- ions are present. But the chloride acts also as competitive inhibitor for the binding of NADH and NAD^+ . From fluorescence-quenching studies, it appears that a Cl^- ion forms a ternary complex with enzyme and NADH (with an association constant of approx. 1.5). The picture is complex, however. NADH is ligated at four additional places (besides the two coenzymes located at the active site) and there is more than one type of Cl^- binding site on the enzyme. A decrease in fluorescence intensity and also a loss of an extrinsic Cotton effect upon the addition of Cl^- ion is explained as due to a conformational alteration in the protein (Coleman & Weiner, 1973 *a, b*).

III. UNIVALENT CATIONS BINDING PROTEINS

Even though the metal binding strength of univalent cations by proteins is considered weak when compared to that of Ca^{2+} or Mg^{2+} and even more so when compared to that of Zn^{2+} , Fe^{2+} , Cu^{2+} , Mo^{2+} , these ions

still appear to be capable of altering the potentially flexible protein conformation.

The role of monovalent cations as modulators of protein structure (allosteric effectors) has been suggested in several instances: Himes & Wilder (1965) showed that crystalline formyltetrahydrofolate synthetase from *Clostridium cylindrosporum* is fully active in the presence of K^+ , NH_4^+ and Rb^+ , partially active in the presence of Cs^+ , and essentially inactive in the presence of Na^+ , Li^+ , or Tris ions. They suggested on the basis of kinetic data, that a conformational change of the proteins occurs in the absence of an activating cation that partially or completely destroys the binding sites of two of the substrates of the reaction, tetrahydrofolate and formate. The enzyme ligates the monovalent cation before accepting these substrates. Scott & Rabinovitz (1967), however, established that the enzyme has a molecular weight of 230 000 (tetramer) in the presence of 50 mM- K^+ , NH_4^+ or Rb^+ and of 58 000 (monomer) in the presence of 50 mM- Na^+ , Li^+ , Tris, Cs^+ or Mg^{2+} . Hence, it is possible that the active site is formed from portions of each subunit when they associate in the presence of certain monovalent ions.

Homoserine dehydrogenase of *Rhodospirillum rubrum* catalyzes reductive conversion of aspartate β -semialdehyde to homoserine. The latter is a precursor of methionine and also, through a separate sequence of reactions, of threonine and then isoleucine. The purified enzyme is subject to feedback inhibition control by L-threonine, an inhibition which is relieved by either L-methionine or L-isoleucine. K^+ ion is required for stability of the dehydrogenase, in both crude and purified preparations. Removal of K^+ causes loss of the sensitivity toward threonine and isoleucine. Subsequent addition of K^+ restores this sensitivity. Alteration in molecular size upon desensitization of the enzyme could not be detected by Sephadex gel filtration (Datta & Gest, 1965).

K^+ ions cause a conformational change resulting in an association of the monomer form of δ -aminolevulinic acid dehydratase of *Rhodospseudomonas spheroids* into an equilibrium mixture of three enzymically active species, a monomer, a dimer, and a trimer. In the absence of cation (0.05 M-KCl) a plot of activity versus substrate changes from the hyperbolic to the sigmoidal type (Nandi, Baker-Cohen & Shemin, 1968; Nandi & Shemin, 1968).

Similarly, in the case of muscle adenosine-5-phosphate deaminase, the enzyme displays hyperbolic kinetics in the presence of K^+ ions

and sigmoidal kinetics in their absence. Na^+ will replace K^+ with equal efficiency. While a 0.1 M ion concentration fully activates the pure enzyme there is no absolute requirement for these ions (Smiley & Suelter, 1967).

In aspartokinase-homoserine dehydrogenase of *Escherichia coli*, a bi-functional enzyme, the allosteric ligands, aspartate and K^+ shift a conformational equilibrium in favour of a catalytically active form whereas threonine binds preferentially to an inactive form. Both enzymic functions are activated by K^+ (but not by Na^+). The co-operativity of the ligand binding can be controlled by the presence or absence of K^+ . When K^+ is lacking, aspartate or threonine have no co-operative effect and there is little change in circular dichroism. When K^+ is added conformational effects are sizable: Ellipticity changes sigmoidally with ligand concentration (Heck & Truffa-Bachi, 1970).

The binding to pyruvate kinase of cationic co-factors (when the non-activator Tris-Cl is included) induces absorbance differences in the wavelength range between 280 and 310 nm. These differences resulting from a comparison of 0.1 M-KCl, NH_4Cl or NaCl – all activator-cation salts – with 0.1 M Tris-Cl are almost identical with those observed in a comparison of the effect of LiCl – a non-activator cation – with another non-activator cation, Tris-Cl. In the case of this enzyme there is no evidence, therefore, that activator cations as opposed to non-activator cations are responsible for perturbation at tryptophan residues, i.e. the tryptophan activation does not explain the activation effect upon enzymic activity. Indeed, pyruvate kinase exhibits the greatest stability in 0.05 M solutions of K^+ , Rb^+ and NH_4^+ , intermediate stability in solutions in which Na^+ is the predominant cation and the least stability in solutions in which Tris^+ and Li^+ predominate. This order of effectiveness of ions in maintaining stability is essentially identical with the order of effectiveness of univalent cations as activators of the pyruvate kinase (Kayne & Suelter, 1965; Wilson, Evans & Becker, 1967).

Here, a word of caution about the interpretation of difference spectra needs to be inserted: If the total difference spectrum characteristic of tryptophan were due to a charge perturbation caused by cation binding, then a very large number of tryptophan residues of pyruvate kinase would have to be involved (Kayne & Suelter, 1965). Actually, perturbation in the solvating environment of the chromophores may be far greater than that in the electrostatic environment. Changes in solvating environment, however, may in turn induce alteration of con-

formation of the protein structure. Andrews & Forster (1972) suggest the following empirical rule to distinguish between solvent-induced and charge-induced spectral perturbations. If the ratio of the molar absorptivity change at 286 nm to that at 292 nm is greater than 0.5, the perturbation is most likely solvent induced, and if the ratio is less than 0.4, the perturbation is most likely charge induced.

K⁺ ions raise the affinity of the pyruvate kinase – Mn²⁺ complex for substrate phosphoenolpyruvate and conversely the phosphoenolpyruvate enhances the affinity of the enzyme-Mn²⁺ complex for K⁺ ions. If phosphoenolpyruvate analogs lacking a free carboxyl group are substituted for the substrate, K⁺ lowers the affinity of the complex for the analogs thus suggesting that the carboxyl group of phosphoenolpyruvate is coordinated by enzyme-bound K⁺ and that this coordination changes the enzyme-Mn²⁺-phosphoenolpyruvate complex to its catalytically active form. It is necessary to wean oneself from the concept that monovalent ions merely stabilize the active conformation of an enzyme. The monovalent ion possibly aids alignment of the phosphoryl group of phosphoenolpyruvate for nucleophilic attack (Nowak & Mildvan, 1972).

The concentration of activator univalent cations necessary for maximum velocity of pyruvate kinase is about 0.05 M. Kachmar & Boyer (1953) showed that K⁺ influences the *V*_{max} but not the *K*_m of this enzyme toward phosphoenolpyruvate. On the other hand, in the case of citrate-synthetase from *E. coli* which shows increased stability towards heat and urea and a shift in the u.v. spectrum in the presence of 0.1 M-K⁺ and other monovalent ions (NH₄⁺ > Na⁺ > Li⁺) the *K*_m for acetyl CoA and oxaloacetate is lowered but maximum velocity remains unchanged (Faloona & Srere, 1969).

Unfortunately, no affinity constants are available in the literature for the binding of univalent ions by the enzymes mentioned above.

IV. CALCIUM BINDING PROTEINS

Muscle contracts by a mechanism in which thin actin-containing and thick myosin-containing filaments slide past each other due to forces induced by cyclically operating cross-bridges. In vertebrate striated muscle interaction between the actin and myosin is regulated by a system present in the actin-containing filaments: the tropomyosin-troponin system. Control of myofibrils, the contractile elements of the

muscle cells, is effected through alterations of the free calcium concentration over a range of 10^{-7} to 10^{-5} M, and the association constant for a protein ion acceptor should be $\sim 10^6$ (M^{-1}). Translocation of calcium from the sarcoplasmic reticulum (a sleeve-like system of vesicles) to the myofibrils in the striated muscle plays a major role in initiating and sustaining contraction and in causing relaxation. Upon stimulation, calcium is released from cellular stores. There is an accumulation of the free ion around the myofibrils before contraction and restoration of calcium levels within certain cellular components before muscular relaxation. Troponin, found by Ebashi & Kodama (1965) to be located at specific points in another protein molecule, tropomyosin, exerts a depression on the reaction of F-actin and myosin in the absence of Ca^{2+} ions. It turns off the active site of actin, preventing interaction with nucleotide-containing myosin and turns it on again after the appearance of calcium, i.e. troponin is believed to play a key role in reversibly alternating contraction and relaxation of actomyosin by responding to small changes in sarcoplasmic calcium concentration. Actually, a troponin 'complex' consists of several proteins: TN-C (or A), an 18000-dalton peptide; TN-I (or B), a 22000-dalton protein and TN-T, a 37000-dalton, tropomyosin-binding, protein. TN-I inhibits actin-myosin interaction independently of calcium. TN-C binds Ca^{2+} and this presumably is the basis of its mechanism of action in releasing the inhibition of the myosin ATPase by the tropomyosin-TN-I complex. When the Ca^{2+} concentration falls to 10^{-7} or less, the calcium binding protein (TN-C) can no longer neutralize the inhibitory protein (TN-I).

Arai & Watanabe (1968) report that 30 μ moles of calcium attach to one gram of unfractionated troponin (i.e. approximately two moles of ion per mole of protein) with an apparent affinity constant of 3.7×10^5 (M^{-1}). According to these workers, precipitation of troponin by calcium and disaggregation of troponin by the reducing agent dithiothreitol support the view that subsequent to calcium binding, troponin undergoes conformation changes which are in turn transmitted through tropomyosin to the actin-myosin interaction. Van Eerd & Kawasaki (1972) claim that ORD measurements at 233 nm suggest an increase of the helix content and fluorescence polarization studies indicate higher rigidity of the Ca^{2+} binding protein in the presence of this ion, but a similar effect is obtained with Mg^{2+} ions (Nagy, Greaser & Gergely, 1973). Saturation of TN-C with calcium ions gives this protein the capacity to bring TN-T into

solution at low ionic strength (at which it would be insoluble otherwise.) Ca^{2+} free TN-C is not effective, i.e. the TN-T-TN-C complex assumes different structures depending on the presence or absence of Ca^{2+} ions (Ebashi, Ohtsuki & Mibashi, 1972). Only troponin TN-C among the separated components displays a high affinity calcium binding site – 47 μmoles attach to 1 gram (i.e. one mole Ca^{2+} /mole of protein) with an apparent affinity constant of $1.4 \times 10^6 \text{ (M}^{-1}\text{)}$. An additional binding site appears when equimolar amounts of TN-C and TN-I are mixed (Greaser *et al.* 1972).

In molluscan muscle, troponin and its components are absent and both the calcium binding and the calcium requirement of the ATPase activity are associated with myosin. About 1.5 μmoles calcium are bound by 1 g protein at free calcium concentrations of $5 \times 10^{-7} \text{ M}$ by the molluscan myosin (while rabbit myosin holds less than 0.2 μmole at that concentration) (Kendrick-Jones, Lehman & Szent-Györgyi, 1970). Such affinity for the ion by the myosin would certainly satisfy the physiological requirement for a constant of the order $10^6 \text{ (M}^{-1}\text{)}$.

In fish and amphibian muscle, calcium binding proteins (M.W. about 12000) not identical with TN-C, acidic in nature, bind 2 moles of Ca^{2+} per mole of protein with an association constant of $10^7 \text{ (M}^{-1}\text{)}$ and 3–6 further moles of the ion albeit 100 times less strongly. The sites for the two tightly ligated ions are chemically indistinguishable (Benzonana, Capony & Pechère, 1972). The conformation of each binding site, however, is different.

Each of the two tightly ligated Ca^{2+} ions is coordinated by six oxygen atoms in an octahedral arrangement. In the CD loop of the protein the carbonyl oxygen of phenylalanine 57 and in the EF loop that of lysine 96 co-ordinates the Ca^{2+} ion. Additional co-ordination is provided in the latter case by the carboxyl oxygen atoms of aspartate residues 90, 92 and 94 and of glutamic acid 101. Aspartate 92 binds the calcium with both of its carboxyl oxygen atoms.

Six of the ten carboxyl oxygens which coordinate the two calcium ions are internal. The EF bound calcium appears accessible to solvent in one direction but no bound water can be located. The CD bound calcium is not exposed to solvent. At each site there are four carboxylate groups and no lysine or arginine residue is near enough to obtain electrical neutrality.

The two metal binding regions are related by an approximate two fold intraprotein axis. While they are 11.9 Å apart, they are connected

by hydrogen bonds involving isoleucines 58 and 97. Thus, cooperativity might exist between these two regions (Kretsinger & Nockolds, 1973).

From examination of the behaviour of tripeptides one may deduce that calcium is coordinated by oxygen and not by nitrogen ligands. This rule is valid in the case of the calcium binding proteins and also in that of thermolysin and that of staphylococcal nuclease.

Ca^{2+} is necessary for the thermostability of thermolysin but not for its enzymic activity. Four such atoms are rather tightly held by the protein molecule (Feder, Garrett & Wildi, 1971). Two Ca^{2+} ions are situated close together with a centre to centre distance of 3.8 Å. They are surrounded by a cluster of ordered water molecules, backbone carbonyl groups and they chelate to the carboxyl groups of three aspartic and two glutamic acid residues. Their coordination structure may be octahedral. A third Ca^{2+} ion located on the surface of the enzyme interacts with two aspartates and a fourth ion, also on the surface, involves an additional aspartate, i.e. even though the inner coordination shell of these four Ca^{2+} ions is probably always octahedral, the backbone configuration provided by the polypeptide is not necessarily identical for each ion (Matthews *et al.* 1972).

In the case of staphylococcal nuclease, the carboxylates of two aspartates (21 and 40) and of a glutamate (43) are clearly complexed to the calcium ion and together with that of a third aspartate (19) (albeit farther away), they ring it in the form of a square. The peptide carbonyl oxygen of threonine (41) is also in this primary coordination sphere but in a displaced octahedral geometry. There is a secondary hydrogen-bonded co-ordination shell built around the Ca^{2+} with the oxygens of threonines, water molecules and other structural elements (Cotton *et al.* 1971).

Using a high resolution spectrometer operating at 100 Mc/s and time-averaging with an internal lock system to minimize drift, it is possible to obtain useful spectra of small proteins. In such proton magnetic resonance spectra, the four histidine C-2 proton peaks of staphylococcal nuclease can be definitely resolved. (The protons on the amino and imidazole nitrogens and carboxyl group exchange rapidly with water and their absorption peaks are not observed.) One histidine C-2 peak of the nuclease is shifted downfield by 16 c/s upon addition of Ca^{2+} above pH 6.8, i.e. to the uncharged imidazole. The positions of the other three peaks are only slightly affected by this ion, less than 6 c/s, possibly due to 'non-specific' interaction (Meadows *et al.* 1967). Besides

the local impact due to the binding of the Ca^{2+} ion, a more general phenomenon can also be observed: Staphylococcal nuclease is stabilized against hydrogen exchange when calcium ions and the inhibitory ligand 3',5'-thymidine diphosphate are present. After EDTA treatment of the enzyme, addition of Ca^{2+} or inhibitory ligand alone stabilizes it to some degree – but to a much lower extent than with the two substances in combination (Schechter, Morávek & Anfinsen, 1968).

Trypsinogen possesses two binding sites for Ca^{2+} . One site has an association constant of $3 \times 10^3 \text{ (M}^{-1}\text{)}$. Upon attachment of the ion, a conformational change is induced which renders the molecule more resistant against formation of inert protein, i.e. the ion has an antagonistic effect on the autolysis of the enzyme and on unmasking of tyrosine residues by exposure to high pH. Furthermore, the ion increases the value of λ_c in ORD suggesting enhanced organisation of the molecule in its presence. (This site also exists in trypsin.) At very high Ca^{2+} concentration an additional site is filled ($K \sim 60$) without evoking structural changes. The second ion is probably located on two of the four adjacent *N*-terminal aspartyl residues which are liberated as $\text{val(asp)}_4\text{lys}$ during the activation and it accelerates the hydrolysis of the lys-ileu bond (Delaage & Lazdunski, 1967; Abita *et al.* 1969).

In the case of liver transglutaminase, Ca^{2+} is necessary for activity (K approx. 1.3×10^2). The essential -SH group of the enzyme reacts with certain thiol reagents only in the presence of the cation. The u.v. difference spectrum (probably involving a contribution from both tryptophan and tyrosine) suggests that the combination of enzyme and Ca^{2+} causes a conformational alteration and the fact that the calcium-activated enzyme is less readily digested by trypsin might be considered confirmatory evidence (Folk, Mullooly & Cole, 1967).

Phospholipases from porcine pancreas and from venom of *Crotalus adamanteus* exhibit an absolute specific requirement for Ca^{2+} . First, the metal ion is ligated to the protein and thereafter the substrate attaches. Alkaline earth cations (Ca^{2+} , Ba^{2+} and Sr^{2+}) when bound by the enzyme isolated from the venom induce spectral perturbations of a tryptophan by removing charged groups in the vicinity of this residue. Zn^{2+} and Cd^{2+} also give rise to spectral changes even though the character of these is somewhat different. Those perturbations accompanying the addition of Ca^{2+} might be primarily due to charge effects while those accompanying the addition of Zn^{2+} might be due to conformational changes. The protein molecule (30000 daltons) binds two ions of Zn^{2+}

or Ca^{2+} with association constants of $1 \times 10^6 \text{ (M}^{-1}\text{)}$ and $1.8 \times 10^4 \text{ (M}^{-1}\text{)}$ respectively. Since there are two subunits, each unit binds one metal ion. Only the Ca^{2+} is needed by this phospholipase for enzymic activity. All the other ions are competitive inhibitors. Possibly, the Ca^{2+} causes the deprotonization of an essential lysine residue at the active site and also induces a conformational alteration which removes the $\epsilon\text{-NH}_2$ group from contact with solvent (Wells, 1973*a, b*).

For full enzymic activity, human salivary amylase requires at least one Ca^{2+} ion per mole and *B. subtilis* amylase (consisting of two subunits) 4–5 ions per mole (Hsiu, Fischer & Stein, 1964), or 2 ions per mole (probably with a K of $4 \times 10^4 \text{ (M}^{-1}\text{)}$) (Levitzki & Reuben, 1973). So far, no evidence has been adduced, however, to show that the metal confers a structural rigidity necessary for catalytic function. Circular dichroic spectra of native enzyme and of Ca^{2+} free enzyme are identical (Smolka, Birnbaum & Darnall, 1971). On the other hand, Ca^{2+} ions are essential if reduced takaamylase A, which in the native state has four disulphide bonds and a molecular weight of 47000 is to refold correctly during reoxidation (Takagi & Isemura, 1965).

Several Ca^{2+} binding proteins (other than phosphoproteins) are being studied at present, e.g. vitamin D_3 , a sterol required by several species for the optimal absorption of Ca^{2+} , leads to the appearance of calcium-binding protein in the mucosa of the small intestine. The association constant for this protein isolated from chickens is reported as approx. $10^6 \text{ (M}^{-1}\text{)}$ and it seems that one mole of protein (28000 daltons) binds four atoms of Ca^{2+} (Wasserman, Corradino & Taylor, 1968). Overall cAMP phosphodiesterase from bovine heart requires a Ca^{2+} binding protein for full enzymic activity. Ca^{2+} attaches to high and low affinity sites with a K of $3 \times 10^5 \text{ (M}^{-1}\text{)}$ where $n = 1$ and a K of $0.8 \times 10^4 \text{ (M}^{-1}\text{)}$ where $n = 2$, respectively. The protein is very acidic and has a mol. weight of about 20000 (Teo & Wang, 1973). Nothing is known, however, about the nature of the bond or about the possible perturbations due to attachment of the Ca^{2+} to the last two mentioned proteins.

In certain instances, a protein will ligate a particular ion after another kind of ion has first been bound in order to induce the appropriate conformation: Concanavalin A accepts Ca^{2+} subsequent to the coordination of a transition metal ion by the amino acid side chains. This protein, a saccharide-binding globulin extracted from *Canavalin ensiformis* (Jack Bean) contains both Mn^{2+} and Ca^{2+} in its native state. At pH below 6.0 the molecule (~ 50000 daltons) is composed of two

identical subunits and contains two of each of the metal binding sites, designated S_1 and S_2 , and two saccharide binding sites. At pH above 6.0 four monomers interact. The K for binding Mn^{2+} is 2×10^3 (M^{-1}) and that for binding Ca^{2+} is 3×10^3 (M^{-1}). On removal of these ions (e.g. by dialysis against 0.1 N HCl) the protein can ligate the transition metal ions, i.e. Mn^{2+} , Co^{2+} , Ni^{2+} , Zn^{2+} and Cd^{2+} at S_1 . Ca^{2+} and Mg^{2+} , however, are not acceptable. On occupation of S_1 the Ca^{2+} binding site, S_2 , is formed. It has equal affinity for Ca^{2+} or Cd^{2+} , little for Sr^{2+} and none for Mg^{2+} , Ba^{2+} , Mn^{2+} and Ni^{2+} ions. This suggests that S_2 is a rigid site which can accommodate only divalent metals of radii approximately 1 Å (Shoham, Kalb & Pecht, 1973). When the two species of metals are added back, in the order indicated, the affinity towards α -glucopyranoside, polysaccharide, etc., is restored. X-ray crystallography studies suggest that at least one of the ligands in S_1 is an imidazole whereas S_2 contains carboxyl ligands only. The Mn^{2+} and Ca^{2+} are 4.3 Å apart and are six and five coordinated, respectively.

The symmetry of the ligands attached to Mn^{2+} is nearly octahedral. From the *N*-terminus the peptide chain runs through Sheet II (the protein contains no helix and each subunit is a six strand β pleated sheet) winding around the Mn^{2+} and Ca^{2+} . While Glu-8 gives the carboxyl group to Mn^{2+} , Asp-10 donates its group to both Mn^{2+} and Ca^{2+} (one oxygen to each metal ion). Tyr-12 supplies its carbonyl oxygen to Ca^{2+} . Asx-14 attaches its side group to Ca^{2+} also. Then the chain loops out into solution and when it returns it lies underneath the metal ions in such a manner that Asp-19 contributes both its carboxyl oxygens – one to each ion – and His gives its imidazole $N^{\epsilon 2}$ to Mn^{2+} . As the chain returns through Sheet II, Glx-34 donates an oxygen or possibly nitrogen to Mn^{2+} . The sixth ligand for Mn^{2+} might come from H_2O and the fifth side chain to Ca^{2+} possibly from Glu-202 (that is, the other end of the chain) (Hardman & Ainsworth, 1972).

Since Mn^{2+} has to be attached to the apoprotein before the Ca^{2+} ion can be ligated, undoubtedly, the Mn^{2+} at least repositions the two amino acids responsible for the 'double site': Asp-10 and Asp-19, so that Ca^{2+} binding can proceed. In addition, ESR studies indicate that the electronic environment of the transition metal-binding site is altered on binding Ca^{2+} ions. Thus, a conformational change is also induced upon ligating the latter ion. An increased intensity of the spectrum probably means that the coordination symmetry of the bound Mn^{2+} is enhanced (Nicolau, Kalb & Yariv, 1969). Only when both the transition

metal and the alkali earth metal are at their places can the carbohydrate be accepted. When the two ions are absent, negatively charged groups concentrated at the potential site probably force that portion of the chain into the solvent. Introduction of the two metal ions moves the region back and stabilizes its conformation. It appears, then, that a small conformational alteration takes place on ligation of transition metal ion and a larger change occurs subsequent to the attachment of the Ca^{2+} . Prima facie evidence for structural alteration derives also from the observation that the apoprotein cannot be crystallized in the same space group as the 'metal-protein complex', ($P2_122_1$ and $I222$, respectively) (Jack, Weinzierl & Kalb, 1971).

V. MANGANESE AND MAGNESIUM BINDING PROTEINS

Pyruvate kinase of muscle requires a divalent cation besides the monovalent one (like K^+) for activation (*vide supra*). (The analogous enzyme of yeast needs only the divalent ion.) Utilizing the effect of the paramagnetic Mn^{2+} or the NMR line width of thallium-205 – which attaches at the K^+ binding site – Kayne & Reuben, (1970) showed that the Mn^{2+} and the Tl^+ are less than 7.4 Å apart, and are presumably both bound at the active site of the enzyme. While the muscle protein is composed of four subunits and has four binding sites for Mn^{2+} , it is possible that an equilibrium mixture of two conformational forms of the enzyme exists: One which has sites for the divalent cation, the other which has none – and that K^+ stabilizes the active form. Measurements of longitudinal proton relaxation rates of water on solutions of the enzyme and Mn^{2+} suggest that three water molecules remain coordinated in the first sphere of the Mn^{2+} bound to the enzyme. Assuming a coordination number of six, a contribution of three links to the metal by the protein appears possible (Reuben & Cohn, 1970).

While addition of Mn^{2+} or Mg^{2+} to pyruvate kinase also produces the u.v. difference spectrum, suggesting perturbation of tryptophan, Co^{2+} does not (Suelter & Melander, 1963). Mg^{2+} or Co^{2+} can replace the Mn^{2+} as activator but Ca^{2+} acts as an inhibitor. The agreement of the kinetic activator constant for Mn^{2+} with its association constant ($1.3 \times 10^4 \text{ (M}^{-1}\text{)}$) indicates that the enzyme acts as metal-protein complex (Mildvan & Cohn, 1965).

Kinetic and magnetic resonance studies sketch the following sequence of events: The divalent ion combines with the protein resulting in

a conformational change. Next, this ion serves as bridge between the enzyme and its nucleotide substrate (ADP). The enzyme – Mn^{2+} – phosphoenolpyruvate complex formation requires an additional change of the environment of the Mn^{2+} other than mere replacement of water ligands (possibly a conformational change in the protein at the binding site) (Mildvan & Cohn, 1966).

The ligation of Mg^{2+} or Mn^{2+} to the first site on yeast enolase (the enzyme possesses two active sites per molecule) causes the appearance of a difference spectrum with peaks at 275, 285 and 295 nm (a tryptophan perturbation spectrum) reflecting changes in the conformation of the protein molecule (Hanlon & Westhead, 1965). Fluorescence measurement suggest that the enzyme becomes more compact (Brewer & Weber, 1966). Actually, two metal ions seem to bind per 67 000 molecular weight unit in the absence of substrate. Intrinsic association constants for Mn^{2+} obtained by equilibrium dialysis as well as gel filtration on Sephadex G-25 are 8×10^5 and 5×10^4 (M^{-1}) for the first and second site, respectively. The activation constant is nearly equal to the dissociation constant at the second site indicating that both sites must be filled for catalytic activity. A tighter bound Mg^{2+} competes for the same site as the weaker bound Mn^{2+} , i.e. these two ions exhibit opposite site preference. Since enolase is composed of two subunits, the large difference in constants for the first two sites would indicate nonidentity of these two units or anticooperative interaction (Hanlon & Westhead, 1969a). [In the presence of substrate (an equilibrium mixture of 2-phosphoglyceric acid and phosphoenolpyruvic acid) at least four Mn^{2+} ions are ligated with the constants 3×10^4 and 2×10^4 (M^{-1}) for the third and fourth ions, respectively (Hanlon & Westhead, 1969a).] Since the kinetically determined activation constant for the metal seems independent of substrate concentration and the K_m values for 2-phosphoglyceric acid and phosphoenolpyruvic acid seems independent of metal ion concentration, a bridge role for the cation was not considered possible, i.e. both metal ion and substrate were expected to bind to the enzyme independently (Hanlon & Westhead, 1969b). Utilizing a wider range of concentration, however, it was found more recently that increasing MnCl_2 concentrations decrease the K_m of 2-phosphoglycerate (Nowak, Mildvan & Kenyon, 1973). Therefore, it is suggested that a ternary enzyme- Mn^{2+} -2-phosphoglycerate bridge complex is formed.

Glutamine synthetase of *E. coli* (mol. weight about 600 000) consists of 12 protomers. The enzyme contains up to 40 equivalents of Mn^{2+}

held with a K of 2.5×10^5 (M^{-1}). While the Mn^{2+} stabilizes the polymeric structure (disaggregation is facilitated by EDTA) the metal is not involved in the linkages that hold the subunits together: Removal of the Mn^{2+} with EDTA results in loss of catalytic activity, an increase in intrinsic viscosity (i.e. a 'relaxed' form of the enzyme is produced) but no dissociation into subunits. Upon readdition of the metal to the undissociated apoenzyme a lag phase associated with the reactivation is observed. This delay, however, can be eliminated by prior incubation of the protein with $0.5 \times 10^{-3} M$ - Mn^{2+} . As the metal is bound, there occurs a rapid exposure of hydrophobic groups (monitored by measuring fluorescence in the presence of 2-p-toluidinylnaphthalene-6-sulphonate), a corresponding burying of the -SH groups and a slower alteration in the environment of aromatic side chains (evidenced by a red shift) (Kingdon, Hubbard & Stradtman, 1968; Shapiro & Ginsburg, 1968).

The bivalent metal ion requirement of the phosphoglucumutase reaction is most effectively satisfied by Mg^{2+} or Ni^{2+} and less so by Co^{2+} and Mn^{2+} and least by Zn^{2+} . The binding constants, however, are about 10^4 for Mg^{2+} , 10^7 for Mn^{2+} , 10^9 for Co^{2+} and 10^{11} (M^{-1}) for Zn^{2+} (Ray, 1969). A metal ion-induced difference spectrum, which is the same for all these bivalent metal ions, involves both tryptophan and tyrosine residues (Peck & Ray, 1969).

For inorganic pyrophosphatase from yeast, two classes of sites for binding of Mg^{2+} , an activator of the enzyme, exist: $n_1 = 1.7$ with a K of 2.5×10^4 (M^{-1}) and $n_2 = 2.2$ with a K of 3.3×10^3 (M^{-1}). Since this enzyme (M.W. 70000) consists of two subunits, presumably each subunit contains one site for each class. The lower affinity sites are activator-binding sites and attachment of metal ions to these induces spectral changes which are similar for Mg^{2+} , Mn^{2+} , Co^{2+} and Zn^{2+} . Interestingly, as in the case of phosphoglucumutase, the order of the metal ions effective in activation seems to be the inverse order of that of their binding strengths. The role of the other two binding sites is unknown (Rapoport *et al.* 1973). Mn^{2+} , Co^{2+} or Zn^{2+} exert a time dependent inhibitory effect. The latter ions induce a conformational change of the protein and in this manner they affect the rate of hydrolysis at the substrate site. Circular dichroism measurements with Mg^{2+} or Mn^{2+} containing enzyme solutions yield slightly different spectra, i.e. every divalent metal ion ligated to the activator site induces its corresponding enzyme conformation. The time dependence, with gradual dissociation or polymerization and slow displacement of one ligand by another ruled

out, strongly suggests a transition of one enzyme conformation to another (Höhne & Rapoport, 1973).

Similarly, time delay has been observed in the activation of β -galactosidase from *E. coli* by Mg^{2+} . While Mg^{2+} and Mn^{2+} activate, Ca^{2+} and Be^{2+} inhibit. The K for Mg^{2+} binding directly to the enzyme is 1.5×10^6 (M^{-1}). The larger the Mg^{2+} concentration, the smaller the lag phase (Tenn, Viratelle & Yon, 1972).

Pyruvate carboxylase is a manganese metalloenzyme in which Mn^{2+} is present in a remarkable nondissociable form (since it remains attached even when exposed to EDTA) in a stoichiometry of 4 moles per mole of enzyme (Scrutton, Utter & Mildvan, 1966). In addition, pyruvate carboxylase from kidney exhibits an absolute requirement for Mg^{2+} as a dissociable ion. Here, as in the case of inorganic pyrophosphatase, the metal ion has two separate functions: it does not only bind to the substrate but it also attaches directly to the enzyme and proceeds to activate the reaction. Mg^{2+} decreases the apparent K_m value for Mg ATP^{2-} suggesting that the cation induces a conformational change. Fluorescence measurements with 1-anilinonaphthalene as a probe, and the observation that the activation is time dependent, reinforce this assumption (Bais & Keech, 1972). A plot of reaction velocity as a function of Mg ATP^{2-} concentration changes from the sigmoidal to the hyperbolic type in the presence of 1×10^{-3} M - $MgCl_2$ (Keech & Barritt, 1967).

Mg^{2+} is required for maximum activity of spinach ribulose-1-5-diphosphate carboxylase. Addition of the metal ion causes a shift in the optimum pH of the enzyme from 8.5 towards neutrality and also provides protection against digestion by Nagarase. In this CO_2 fixation, reaction a plot of product formation as a function of bicarbonate concentration is sigmoidal in character (Sugiyama, Nakayama & Akazawa, 1968). From the data it appears to us that addition of Mg^{2+} changes this curve to the hyperbolic type (even though the authors claim that the interaction coefficient remains constant).

For fructose 1-6 diphosphatase from liver a constant of 2.3×10^4 (M^{-1}) for the direct attachment of Mn^{2+} to the protein has been reported. This enzyme has a molecular weight of 150000 and possesses four binding sites (Kolb & Kolb, 1973). Although Mn^{2+} is ligated to the enzyme at pH 7.5 in the absence of substrate, at pH 5.8 binding occurs only in its presence (Grazi, Accorsi & Poutiemoli, 1971).

Phosphofructokinase which requires divalent ions for activity contains

1 or 2 Mn^{2+} binding sites with a K of $5 \times 10^4 \text{ (M}^{-1}\text{)}$ and possibly 14 sites with a K of $1 \times 10^3 \text{ (M}^{-1}\text{)}$, all of which can ligate the metal directly (Cottam & Uydea, 1973). No conformational changes subsequent to addition of metal ion to the last mentioned two enzymes have been described so far.

VI. ZINC BINDING PROTEINS

Imidazole groups appear to be primary sites for binding of Zn^{2+} . The maximum co-ordination number for this metal usually is four and the configuration is of the tetrahedral type. The values for the proteins listed in Table 2 indicate that Zn^{2+} binds to relatively high energy sites unlike the alkali and alkaline earth metal cations and also unlike Mn^{2+} (which has no crystal field stabilization energy).

In carbonic anhydrase a 15 Å deep conical cavity, the active site, is built from a Zn^{2+} ion with part of a twisted pleated sheet and some segments in the top of the molecule. The metal is coordinated in a distorted tetrahedral arrangement by three histidine residues (from the central pleated sheet) $\text{N}^{\delta 2}$ of His 93 and His 95, $\text{N}^{\delta 1}$ of His 117 and by one water molecule or hydroxyl ion. Other divalent first transition and II B series metal ions such as Cu^{2+} , Co^{2+} , Mn^{2+} can be bound at the same site instead. (When Hg^{2+} is employed, however, the binding site is moved by about 0.6 Å.) While carbonic anhydrase displays activity when Co^{2+} is attached to its protein, Mn^{2+} , Cu^{2+} or Mg^{2+} complexes are inactive (Kannan *et al.* 1971). At present, there is little direct evidence for identifying the catalytic and substrate binding groups in the enzyme.

Zn^{2+} is the preferred metal ion for carboxypeptidase A. It is bound to $\text{N}^{\delta 1}$ of the imidazole group of His 69, and of His 169 as well as to oxygen of the carboxyl group of Glu 72 and to a water molecule. While the metal appears to have no pivotal role in maintaining the three-dimensional structure of the polypeptide backbone of the enzyme, the oxygen in the carbonyl group of the peptide substrate binds to the Zn^{2+} , displacing the water and introducing strain. The metal acts as a Lewis acid in accepting electrons from the oxygen (Quiocho & Lipscomb, 1971).

If the Zn^{2+} is replaced by different metal ions with an exhibited affinity scale: $\text{Hg}^{2+} \gg \text{Cd}^{2+} > \text{Zn}^{2+} > \text{Ni}^{2+} > \text{Co}^{2+} > \text{Mn}^{2+}$, kinetic parameters are affected differently for various substrates (Coleman &

TABLE 2. *Intrinsic association constants of various proteins for zinc*

Ligand	<i>K</i>
Aspartate Transcarbamylase†	1×10^{16}
Alk. phosphatase‡	$1 \times 10^8, 1 \times 10^{10}$
Carbonic anhydrase*	1×10^{16}
Carboxypeptidase A§	2×10^8
Carboxypeptidase B¶	7×10^5
Enolase††	2×10^5
Insulin‡‡	$4.7 \times 10^6, 3.5 \times 10^4$
Human serum albumin**	1×10^7

† Rosenbusch & Weber (1971).

‡ Cohen & Wilson (1966).

* Lindskog & Nyman (1964).

§ Coleman & Vallee (1960).

¶ Zisapel & Sokolovsky (1973).

†† Malström (1961).

‡‡ Summerell, Osmand & Howard-Smith (1965).

** Giroux & Henkin (1972).

Vallee, 1961). This observation is also valid for carboxypeptidase B (Zisapel & Sokolovsky, 1973). One possible explanation would be that metals alter the micro-environment of the active site.

As in carboxypeptidase A the Zn^{2+} in thermolysin requires two histidines (residues 142 and 146) and one glutamic acid (residue 166) as its ligands, i.e., these two enzymes have this common feature even though their tertiary structures are different. If one approaches the front of the cleft in thermolysin, the Zn^{2+} ligands are below and behind the metal and are arranged approximately as a tetrahedron (Matthews *et al.* 1972). Whether the Zn^{2+} ion in carbonic anhydrase, carboxypeptidase A and thermolysin has other functions besides its participation in the catalytic mechanism itself remains to be seen. (The role of Ca^{2+} in thermolysin is described under Calcium Binding Proteins.)

While alkaline phosphatase isolated from *E. coli* consists of two identical subunits (each of molecular weight 43 000) and contains two Zn^{2+} ions per enzyme molecule, it is possible that it only possesses one binding site for the substrate. The enzymic activity of the one-zinc form is 0.12 compared to the regular two-zinc form (Cohen & Wilson, 1966). X-ray data show a small region of electron density located 10 Å away from the zinc which shifts by 8 Å when the metal is removed (Knox & Wyckoff, 1973) suggesting that the metal not only plays a role in the catalytic activity of the molecule but also serves to stabilize its

tertiary structure. That the latter is altered by the metal binding had previously been adduced by studies employing u.v. difference spectra and circular dichroism in the 260–320 nm region as well as by the observation that removal of metal ions leads to exposure to solvent of five tyrosine residues (Reynolds & Schlesinger, 1969). From hydrogen ion equilibrium measurement it appears that each ligated Zn^{2+} has at least two protein binding sites, one of which is a histidine residue (Reynolds & Schlesinger, 1968). It will be interesting to see whether X-ray analysis will confirm here again the involvement of this basic amino acid residue as it has done for the other Zn^{2+} containing enzymes. It should be mentioned also that an 8–10 Å hole exists on the dimer axis between the Zn^{2+} ions which might represent the binding site for the substrate (Knox & Wyckoff, 1973).

While zinc-free insulin is able to dimerize, the metal is required for crystal lattice formation. A unit cell is composed of three dimers and two zinc atoms. Hence, the smallest content of Zn^{2+} necessary for crystallization of bovine insulin is 2 atoms per hexamer. These two atoms are tightly bound and cannot be removed by dialysis. The first Zn^{2+} could join together three previously detached dimers, inducing changes in geometry and thus facilitating the binding of the second ion. Utilizing hydrogen ion titration data it was suggested that three imidazole atoms are involved per Zn^{2+} atom and that the metal ion is trigonally co-ordinated (Brill & Venable, 1967). From X-ray data, however, it is apparent that the two zinc ions lie on the threefold axis (which relates the three dimers) approximately 8.5 Å above and below the local axis and that each of the two ions is coordinately held by an imidazole nitrogen ($\text{N}^{\epsilon 2}$) of one of the B 10 imidazoles of each of the dimers. An unusual octahedral arrangement rather than the usual tetrahedral one is indicated, and the remaining bonds are probably filled by water (Blundell *et al.* 1971). As they occur in rhombohedral crystals, the sites for the two Zn^{2+} ions are not identical: The two paths of water structure seen around each metal ion also vary.

It is interesting that insulin isolated from guinea-pigs contains no B 10 histidine and that it does neither bind Zn^{2+} nor associate to dimers or hexamers but that it still exhibits biological activity (Zimmerman, Kells & Yip, 1972). Evidently, the metal is not necessary for the hormonal function of the protein.

Other instances can be cited allowing the conclusion that Zn^{2+} has the general function of cross-linking subunits: Zn^{2+} is essential for the

crystallization of α -amylase isolated from *B. subtilis*. In addition to ligating Ca^{2+} like the other amylases (*vide supra*) this enzyme also contains one ion of Zn^{2+} per enzyme molecule (96 000 daltons). Upon treatment with EDTA the Zn^{2+} is removed and two subunits are formed (Robyt & Ackerman, 1973).

Similar examples may be chosen from the NAD^+ dependent dehydrogenases. E.g. horse liver alcohol dehydrogenase: There are about 3.5 ions of Zn^{2+} attached per enzyme molecule (80 000 daltons). Two of these, the more reactive metal ions towards chelating agents, seem essential for catalytic activity while the other two 'buried' ions appear to play a role in holding subunits together (Drum *et al.* 1967).†

Aspartate transcarbamylase of *E. coli* (M.W. 300 000) is composed of 6 subunits (17 000 daltons) (which form trimers) possessing catalytic sites and 6 subunits (33 000 daltons) (which form dimers) containing regulatory sites. While the enzyme has no metal requirement for catalytic activity, divalent metal ions (Zn^{2+}) are required for association of both these units as well as for stabilization of the regulatory ones. The high affinity sites which form the metal protein complex could provide the thermodynamic driving force for the association and stabilization. The Zn^{2+} is not available to chelating agents such as EDTA and it is estimated that the apparent association constant is either of the same magnitude as that of the Zn^{2+} EDTA complex (about $1 \times 10^{16} \text{ M}^{-1}$) or greater. It is possible, of course, that the EDTA is prevented by steric hindrances from approaching the appropriate surface of the enzyme. At any rate, the lack of equilibration of the enzyme metal complex with chelators precludes an experimental estimate of its thermodynamic stability. The six essential metal ions are located in the six regulatory polypeptide chains. But the Zn^{2+} binding sites in the complete enzyme molecule contain ligating groups from both subunits. Since the Zn^{2+} is quantitatively released from the protein upon substitution of its -SH groups, e.g. by p-hydroxy-mercury benzoate, it is suggested that mercaptide containing residues are ligands to the metal ions (Rosenbusch & Weber, 1971). Whereas the regulatory subunits containing Zn^{2+} are stable dimers in solution, in the absence of the metal

† In yeast alcohol dehydrogenase, an enzyme composed of 4 subunits, 4 zinc atoms are ligated per molecule. Loss of metal is accompanied by a shift in mol. weight from 150 000 to 36 000 (Snodgrass, Vallee & Hoch, 1960). Liver alcohol dehydrogenase, however, is dimeric. In contrast, lactic dehydrogenase and glyceraldehyde-3-phosphate dehydrogenase are devoid of zinc even though they are also tetrameric and have a mol. weight of about 150 000.

ions, monomers are in evidence. Zn-regulatory and apo-regulatory subunits differ in their circular dichroism spectra, suggesting that the addition of the metal induces conformational readjustments in the regulatory chains (Cohlberg, Pigiet & Schachman, 1972).

The Cd^{2+} -enzyme complex has a strong u.v. absorption bond that is absent in the Zn^{2+} derivative. Utilizing Cd^{2+} instead of Zn^{2+} it could be shown that circular dichroism differences (between 250 and 280 nm) accompany the metal replacement and that additional changes in the circular dichroism pattern of the Cd^{2+} chromophore occur upon binding of the substrate carbamyl phosphate, and the substrate analogue, succinate, concomitant with alterations in kinetic and inhibitor binding properties. Hence, conformational alterations induced at the metal binding sites of the regulatory subunits due to the ligation of the Cd^{2+} (which might arise from structural differences within about 10 Å radius of the Cd^{2+} ion) affect the conformation of the active sites on the catalytic subunits (Griffin *et al.* 1973).

A similar tight attachment of Zn^{2+} or Cd^{2+} exists in metallothionein, a protein capable of binding these two ions, as well as Cu^{2+} . Its name refers to the unusual proportion of cysteine residues (26) in the molecule (M.W. ~ 10000). Thus, it may carry 8–9 metal ions per mole, each of which would be complexed with 3-SH groups. Replacement of the Zn^{2+} with Cd^{2+} results in an intense Cotton effect near 255 nm due to the asymmetric binding of the Cd^{2+} to multiple sulphur ligands (Pulido, Kagi & Vallee, 1966). When hydrogen isotope exchange for the apoprotein is compared to that of the metal-protein complex, it is found that the former loses all its isotopic hydrogen upon one passage through a gel filtration column while the latter retains more than 20 hydrogens, indicating that metal attachment introduces occlusion of a portion of the molecule (Ulmer, 1970).

VII. CONCLUSION

The biological environment of protein contains a great variety of inorganic ions which influence the conformation, stability and function of the proteins. These ions may merely exert a non-specific 'long distance' effect by way of an electric field or they may directly and reversibly interact with the protein forming electrovalent bonds. In this review, we address ourselves to the latter phenomenon. We are concerned with inorganic ions that bind directly to the proteins and not

those that form complexes with substrates only. Diversity among proteins in their tendency to combine with these ions, suggests that particular configurations produce specific reactive sites. In turn, binding of ions to these sites alters the electronic and therefore, the local steric conformation of the protein. There is sufficient evidence to deduce that inorganic ions cause changes in the conformation of proteins as do substrates and allosteric effectors even if these ions exhibit a relatively low affinity for the protein, e.g. Ca^+ for liver transglutaminase.

The evidence is adduced by a variety of techniques: X-ray crystallography, nuclear magnetic resonance, optical rotatory dispersion, circular dichroism, spectrophotometric perturbation, fluorescence intensity measurement, employment of fluorescent dye probes, enzyme kinetics, hydrogen isotope exchange, heat stability behaviour, proteolytic degradation.

Perutz (1970) suggested that a small movement (less than 1 Å) of iron in relation to the porphyrin plane in haemoglobin when it goes from a coordination number of 5 to one of 6 is sufficient to move a proximal histidine towards the plane causing further realignment of helix *F* which moves towards the centre of the molecule and expels tryosine HC2 (140) which in turn pulls arginine HC3 (141) with it, etc. That is, the small effect is translated over long distances of the protein molecule. In the case of δ aminolevulinic acid dehydratase from bacteria and adenosine-5-phosphate deaminase from muscle, the presence of potassium ions changes an activity versus substrate plot from the sigmoidal to the hyperbolic type. An identical phenomenon takes place in the pyruvate carboxylase (from kidney) reaction and in the ribulose-1,5-diphosphate carboxylase (from spinach leaves) reaction when Mg^{2+} is added. In these instances, it appears that the metal ions remove a 'constraining' cooperative effect that is imposed when these ions are not ligated to the proteins. Negatively charged groupings neighbouring the point of attachment of metal ions could be displaced, stabilization by cross-linking might also be induced, and by a 'domino effect' protein structure further away could be altered.

It is of interest that a lag phase accompanies the inhibitory effect of divalent ions on pyruvate kinase isolated from muscle and on inorganic pyrophosphatase from yeast, and the activation effect on β -galactosidase and also glutamine synthetase obtained from bacteria as well as pyruvate carboxylase from kidney, strongly suggesting a relatively slow transition of one enzyme conformation to another.

From X-ray analysis it is convincingly apparent that sulphate ions are held by the sidechains of basic amino acids, that carboxyl and carbonyl oxygens ligate Ca^{2+} and that Zn^{2+} prefers imidazole residues but also utilizes carboxyl oxygens for its ligands. Other evidence shows that mercaptide groups might also conjugate Zn^{2+} tightly. What are the 'domino effects' on various portions of the protein structure when the metal attaches to these groups? How is the metal-induced 'lag phase' phenomenon explained?

Of particular further attraction should be studies on the 'induced fit' effect by one class of metal ion as a condition for the binding of a second metal ion where the specificities of the two sites (or double site) are different, e.g. in the case of concanavalin A. (Even though on a basis of affinity to the metal, concanavalin A is labelled a 'metal-protein complex' rather than a 'metallo-protein', the consequences of the metal co-ordination on the structural orientation of the molecule and thereby on its functional behaviour are extremely consequential.)

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